

```

--- Hyperparameters ---
InitialLearnRate : 0.000042
ValidationPatience : 23
L2Regularization : 0.000014
LearnRateDropPeriod : 37
LearnRateDropFactor : 0.41

```

```

--- Performance Metrics(1) ---

```

Set	RMSE	MAE	R ²
Train	0.0231	0.0161	0.9181
Val	0.0262	0.0179	0.8609
Test	0.0190	0.0134	0.9448
Elapsed Time	: 51.37 seconds		

```

rmseVal = single
0.0230

```

Iter	Eval result	Objective	Objective runtime	BestSoFar (observed)	BestSoFar (estim.)	InitialLearnRate	ValidationPatience
1	Best	0.023044	249.11	0.023044	0.023044	4.2465e-05	

```

--- Hyperparameters ---
InitialLearnRate : 0.000967
ValidationPatience : 44
L2Regularization : 0.000003
LearnRateDropPeriod : 90
LearnRateDropFactor : 0.56

```

```

--- Performance Metrics(1) ---

```

Set	RMSE	MAE	R ²
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```

rmseVal = single
0.0071

```

2	Best	0.007078	252.05	0.007078	0.0080307	0.00096678
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```

--- Hyperparameters ---
InitialLearnRate : 0.001298
ValidationPatience : 72
L2Regularization : 0.000239
LearnRateDropPeriod : 33
LearnRateDropFactor : 0.51

```

```

--- Performance Metrics(1) ---

```

Set	RMSE	MAE	R ²
Train	0.0260	0.0185	0.8965
Val	0.0280	0.0192	0.8416
Test	0.0226	0.0158	0.9225

```

rmseVal = single
0.0271

```

| 3 | Accept | 0.027103 | 114.43 | 0.007078 | 0.0070793 | 0.0012984 |

--- Hyperparameters ---

InitialLearnRate : 0.000491
ValidationPatience : 51
L2Regularization : 0.008654
LearnRateDropPeriod : 88
LearnRateDropFactor : 0.54

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0761	0.0640	0.1126
Val	0.0676	0.0579	0.0743
Test	0.0756	0.0598	0.1301

rmseVal = single
0.0753

| 4 | Accept | 0.075312 | 90.562 | 0.007078 | 0.0070835 | 0.00049071 |

--- Hyperparameters ---

InitialLearnRate : 0.009428
ValidationPatience : 53
L2Regularization : 0.000001
LearnRateDropPeriod : 30
LearnRateDropFactor : 0.43

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0073	0.0054	0.9917
Val	0.0089	0.0067	0.9840
Test	0.0078	0.0057	0.9906

rmseVal = single
0.0077

| 5 | Accept | 0.0077219 | 223.84 | 0.007078 | 0.007094 | 0.0094285 |

--- Hyperparameters ---

InitialLearnRate : 0.000227
ValidationPatience : 80
L2Regularization : 0.000006
LearnRateDropPeriod : 42
LearnRateDropFactor : 0.86

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0114	0.0073	0.9803
Val	0.0141	0.0086	0.9595
Test	0.0091	0.0064	0.9873

rmseVal = single
0.0117

| 6 | Accept | 0.011691 | 244.16 | 0.007078 | 0.0070794 | 0.00022709 |

--- Hyperparameters ---

InitialLearnRate : 0.009919

ValidationPatience : 67
L2Regularization : 0.000009
LearnRateDropPeriod : 33
LearnRateDropFactor : 0.77

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0096	0.0071	0.9858
Val	0.0117	0.0083	0.9725
Test	0.0117	0.0077	0.9929

rmseVal = single

0.0116

| 7 | Accept | 0.011558 | 190.71 | 0.007078 | 0.0070715 | 0.0099191 |

--- Hyperparameters ---

InitialLearnRate : 0.009698
ValidationPatience : 59
L2Regularization : 0.000002
LearnRateDropPeriod : 49
LearnRateDropFactor : 0.89

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0077	0.0057	0.9908
Val	0.0094	0.0068	0.9820
Test	0.0078	0.0059	0.9906

rmseVal = single

0.0093

| 8 | Accept | 0.0093475 | 190.83 | 0.007078 | 0.0071149 | 0.0096985 |

--- Hyperparameters ---

InitialLearnRate : 0.000084
ValidationPatience : 18
L2Regularization : 0.000001
LearnRateDropPeriod : 99
LearnRateDropFactor : 0.67

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0174	0.0118	0.9536
Val	0.0207	0.0134	0.9135
Test	0.0128	0.0091	0.9752

rmseVal = single

0.0177

| 9 | Accept | 0.017669 | 248.59 | 0.007078 | 0.0071046 | 8.3912e-05 |

--- Hyperparameters ---

InitialLearnRate : 0.000130
ValidationPatience : 43
L2Regularization : 0.000001
LearnRateDropPeriod : 89
LearnRateDropFactor : 0.88

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0141	0.0092	0.9695
Val	0.0173	0.0107	0.9393
Test	0.0105	0.0074	0.9831

rmseVal = single
0.0143

| 10 | Accept | 0.014346 | 244.37 | 0.007078 | 0.0070906 | 0.00013043 |

--- Hyperparameters ---

InitialLearnRate : 0.000095
ValidationPatience : 69
L2Regularization : 0.000003
LearnRateDropPeriod : 35
LearnRateDropFactor : 0.11

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0166	0.0112	0.9576
Val	0.0199	0.0128	0.9197
Test	0.0122	0.0087	0.9774

rmseVal = single
0.0169

| 11 | Accept | 0.016852 | 245.65 | 0.007078 | 0.0070573 | 9.4925e-05 |

--- Hyperparameters ---

InitialLearnRate : 0.002050
ValidationPatience : 34
L2Regularization : 0.000001
LearnRateDropPeriod : 30
LearnRateDropFactor : 0.87

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0063	0.0046	0.9938
Val	0.0080	0.0060	0.9870
Test	0.0071	0.0053	0.9923

rmseVal = single
0.0065

| 12 | Best | 0.0065457 | 213.03 | 0.0065457 | 0.0064892 | 0.0020504 |

--- Hyperparameters ---

InitialLearnRate : 0.001498
ValidationPatience : 72
L2Regularization : 0.000001
LearnRateDropPeriod : 100
LearnRateDropFactor : 0.84

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0057	0.0039	0.9951
Val	0.0070	0.0047	0.9901

Test 0.0061 0.0043 0.9943

rmseVal = single

0.0063

| 13 | Best | 0.0063082 | 242.63 | 0.0063082 | 0.0063872 | 0.0014983 |

--- Hyperparameters ---

InitialLearnRate : 0.001505

ValidationPatience : 34

L2Regularization : 0.000001

LearnRateDropPeriod : 30

LearnRateDropFactor : 0.87

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0063	0.0044	0.9940
Val	0.0074	0.0052	0.9890
Test	0.0067	0.0046	0.9932

rmseVal = single

0.0066

| 14 | Accept | 0.0065912 | 240.48 | 0.0063082 | 0.0063965 | 0.0015047 |

--- Hyperparameters ---

InitialLearnRate : 0.002241

ValidationPatience : 20

L2Regularization : 0.000001

LearnRateDropPeriod : 95

LearnRateDropFactor : 0.35

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0058	0.0039	0.9949
Val	0.0069	0.0048	0.9903
Test	0.0057	0.0040	0.9950

rmseVal = single

0.0073

| 15 | Accept | 0.0072818 | 167.28 | 0.0063082 | 0.0063415 | 0.0022414 |

--- Hyperparameters ---

InitialLearnRate : 0.001821

ValidationPatience : 76

L2Regularization : 0.000003

LearnRateDropPeriod : 58

LearnRateDropFactor : 0.83

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0059	0.0041	0.9946
Val	0.0076	0.0050	0.9884
Test	0.0062	0.0044	0.9941

rmseVal = single

0.0074

| 16 | Accept | 0.0073807 | 244.63 | 0.0063082 | 0.006426 | 0.0018212 |

--- Hyperparameters ---

InitialLearnRate : 0.000969
ValidationPatience : 67
L2Regularization : 0.000001
LearnRateDropPeriod : 59
LearnRateDropFactor : 0.89

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0055	0.0039	0.9954
Val	0.0067	0.0046	0.9908
Test	0.0056	0.0038	0.9953

rmseVal = single
0.0068

| 17 | Accept | 0.006755 | 244.89 | 0.0063082 | 0.0065127 | 0.00096897 |

--- Hyperparameters ---

InitialLearnRate : 0.001505
ValidationPatience : 34
L2Regularization : 0.000001
LearnRateDropPeriod : 30
LearnRateDropFactor : 0.87

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0063	0.0044	0.9940
Val	0.0074	0.0052	0.9890
Test	0.0067	0.0046	0.9932

rmseVal = single
0.0066

| 18 | Accept | 0.0065912 | 248.35 | 0.0063082 | 0.0065091 | 0.0015047 |

--- Hyperparameters ---

InitialLearnRate : 0.001220
ValidationPatience : 65
L2Regularization : 0.000001
LearnRateDropPeriod : 32
LearnRateDropFactor : 0.87

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0055	0.0038	0.9954
Val	0.0068	0.0047	0.9905
Test	0.0057	0.0040	0.9951

rmseVal = single
0.0066

| 19 | Accept | 0.0066121 | 241.49 | 0.0063082 | 0.0065276 | 0.0012202 |

--- Hyperparameters ---

InitialLearnRate : 0.000010

```
ValidationPatience : 10
L2Regularization : 0.008837
LearnRateDropPeriod : 98
LearnRateDropFactor : 0.81
```

```
--- Performance Metrics(1) ---
```

Set	RMSE	MAE	R ²
Train	0.1143	0.0966	-0.9983
Val	0.1150	0.0963	-1.6785
Test	0.1055	0.0959	0.6012

rmseVal = single
0.1079

	20		Accept		0.10786		245.91		0.0063082		0.0065323		1.0015e-05	
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--- Hyperparameters ---

```
InitialLearnRate      : 0.009847
ValidationPatience    : 74
L2Regularization      : 0.009961
LearnRateDropPeriod   : 63
LearnRateDropFactor    : 0.83
```

```
--- Performance Metrics(1) ---
```

Set	RMSE	MAE	R ²
Train	0.0809	0.0660	-0.0025
Val	0.0703	0.0591	-0.0005
Test	0.0825	0.0640	-0.0355

rmseVal = single
0.0804

Iter	Eval result	Objective	Objective runtime	BestSoFar (observed)	BestSoFar (estim.)	InitialLearn- Rate	Validation tience
21	Accept	0.080391	77.841	0.0063082	0.006545	0.0098474	

--- Hyperparameters ---

```
InitialLearnRate      : 0.000010
ValidationPatience    : 68
L2Regularization      : 0.000253
LearnRateDropPeriod   : 78
LearnRateDropFactor    : 0.75
```

```
--- Performance Metrics(1) ---
```

Set	RMSE	MAE	R ²
rmseVal = single			
0.0542			

22	Accept	0.054173	243.4	0.0063082	0.0065453	1.0103e-05
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```
--- Hyperparameters ---
```

```
InitialLearnRate      : 0.000010
ValidationPatience    : 67
L2Regularization      : 0.000001
LearnRateDropPeriod   : 56
LearnRateDropFactor    : 0.32
```

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0622	0.0467	0.4079
Val	0.0600	0.0459	0.2705
Test	0.0639	0.0447	0.3771

rmseVal = single
0.0524

| 23 | Accept | 0.052423 | 245.74 | 0.0063082 | 0.0065588 | 1.0056e-05 |

--- Hyperparameters ---

InitialLearnRate : 0.000969
ValidationPatience : 67
L2Regularization : 0.000001
LearnRateDropPeriod : 59
LearnRateDropFactor : 0.89

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0055	0.0039	0.9954
Val	0.0067	0.0046	0.9908
Test	0.0056	0.0038	0.9953

rmseVal = single
0.0068

| 24 | Accept | 0.006755 | 242.36 | 0.0063082 | 0.0065614 | 0.00096897 |

--- Hyperparameters ---

InitialLearnRate : 0.001052
ValidationPatience : 54
L2Regularization : 0.000001
LearnRateDropPeriod : 99
LearnRateDropFactor : 0.59

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0058	0.0040	0.9949
Val	0.0072	0.0048	0.9894
Test	0.0059	0.0041	0.9947

rmseVal = single
0.0068

| 25 | Accept | 0.006801 | 241.59 | 0.0063082 | 0.0065677 | 0.0010522 |

--- Hyperparameters ---

InitialLearnRate : 0.000088
ValidationPatience : 58
L2Regularization : 0.000523
LearnRateDropPeriod : 82
LearnRateDropFactor : 0.34

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0285	0.0207	0.8760
Val	0.0291	0.0206	0.8281

Test 0.0251 0.0178 0.9038

rmseVal = single

0.0291

| 26 | Accept | 0.029139 | 240.96 | 0.0063082 | 0.0065702 | 8.8217e-05 |

--- Hyperparameters ---

InitialLearnRate : 0.001314

ValidationPatience : 24

L2Regularization : 0.000001

LearnRateDropPeriod : 84

LearnRateDropFactor : 0.31

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0054	0.0036	0.9955
Val	0.0067	0.0045	0.9909
Test	0.0055	0.0037	0.9954

rmseVal = single

0.0069

| 27 | Accept | 0.0068741 | 236.11 | 0.0063082 | 0.0065678 | 0.0013145 |

--- Hyperparameters ---

InitialLearnRate : 0.009991

ValidationPatience : 57

L2Regularization : 0.000486

LearnRateDropPeriod : 47

LearnRateDropFactor : 0.86

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0329	0.0245	0.8339
Val	0.0334	0.0250	0.7741
Test	0.0297	0.0223	0.8656

rmseVal = single

0.0328

| 28 | Accept | 0.032816 | 94.042 | 0.0063082 | 0.0065526 | 0.0099905 |

--- Hyperparameters ---

InitialLearnRate : 0.000237

ValidationPatience : 74

L2Regularization : 0.000066

LearnRateDropPeriod : 72

LearnRateDropFactor : 0.20

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0164	0.0111	0.9588
Val	0.0198	0.0129	0.9207
Test	0.0131	0.0093	0.9740

rmseVal = single

0.0177

| 29 | Accept | 0.017726 | 243.45 | 0.0063082 | 0.0065412 | 0.00023738 |

--- Hyperparameters ---

InitialLearnRate : 0.000059
ValidationPatience : 78
L2Regularization : 0.009878
LearnRateDropPeriod : 27
LearnRateDropFactor : 0.37

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0674	0.0566	0.3045
Val	0.0638	0.0537	0.1744
Test	0.0639	0.0502	0.3789

rmseVal = single
0.0683

| 30 | Accept | 0.068253 | 245.03 | 0.0063082 | 0.006532 | 5.8578e-05 |

Optimization completed.
MaxObjectiveEvaluations of 30 reached.
Total function evaluations: 30
Total elapsed time: 6507.8157 seconds
Total objective function evaluation time: 6493.5061

Best observed feasible point:

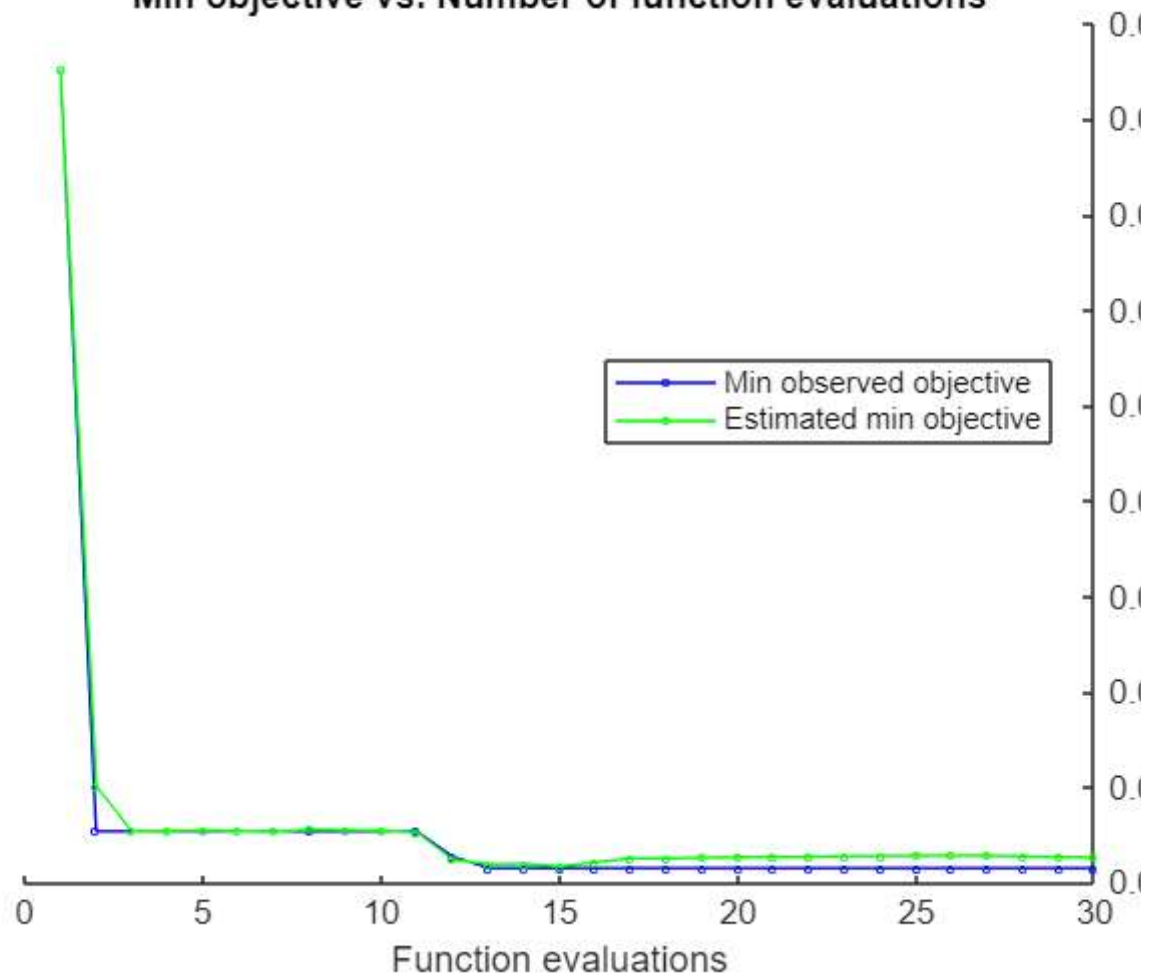
InitialLearnRate	ValidationPatience	L2Regularization	LearnRateDropPeriod	LearnRate
0.0014983	72	1.0055e-06	100	0.84

Observed objective function value = 0.0063082
Estimated objective function value = 0.0065434
Function evaluation time = 242.6335

Best estimated feasible point (according to models):

InitialLearnRate	ValidationPatience	L2Regularization	LearnRateDropPeriod	LearnRate
0.0015047	34	1.0111e-06	30	0.85

Min objective vs. Number of function evaluations



bestParams = 1x5 table

	InitialLearnRate	ValidationPatience	L2Regularization	LearnRate
1	0.0015	72	1.0055e-06	

--- Hyperparameters ---

InitialLearnRate : 0.001498
 ValidationPatience : 72
 L2Regularization : 0.000001
 LearnRateDropPeriod : 100
 LearnRateDropFactor : 0.84

--- Performance Metrics(1) ---

Set	RMSE	MAE	R ²
Train	0.0057	0.0039	0.9951
Val	0.0070	0.0047	0.9901
Test	0.0061	0.0043	0.9943

Elapsed Time : 55.07 seconds

rmseVal = single
 0.0063

0.0063